Explanation of fseek(file, -sizeof(Student), SEEK_CUR) in C/C++

The line of code `fseek(file, -sizeof(Student), SEEK_CUR)` is used in C/C++ for file manipulation, specifically to move the file pointer to a different position in the file. Let's break it down in detail:

1. `fseek(file, -sizeof(Student), SEEK_CUR)`:

   - `fseek`: This function is used to move the file pointer associated with a file stream (`file`) to a specific location within the file. It takes three arguments:

     1. `file`: The file pointer, typically obtained by using `fopen()`.

     2. `offset`: The number of bytes to move the file pointer. This can be a positive or negative integer.

     3. `origin`: The position from where the offset is applied. It can be:

        - `SEEK_SET`: Beginning of the file.

        - `SEEK_CUR`: Current position of the file pointer.

        - `SEEK_END`: End of the file.

2. `file`:

   - This is the file stream pointer, which points to an open file. It's typically obtained from the `fopen` function.

3. `-sizeof(Student)`:

   - `sizeof(Student)`: This function returns the size (in bytes) of the structure `Student`. The size is determined by the sum of the sizes of all the data members within the structure and any padding added by the compiler.

   - `-sizeof(Student)`: The negative sign before `sizeof(Student)` indicates that we want to move the file pointer backward by the size of the `Student` structure.

4. `SEEK_CUR`:

   - This tells `fseek` to move the file pointer relative to its current position.

   - The current position is where the file pointer is before the `fseek` function is called.

Explanation in Context:

Imagine you're reading a binary file that contains records of a `Student` structure. If you have just read one `Student` record and realize that you need to go back to the beginning of that record (maybe to read it again or overwrite it), you can use this `fseek` function to move the file pointer backward by the size of one `Student` structure.

- Scenario:

   - Suppose the file pointer is currently at position `N`.

   - `sizeof(Student)` is, say, 20 bytes.

   - The statement `fseek(file, -sizeof(Student), SEEK_CUR)` will move the file pointer from position `N` to `N - 20`, effectively pointing to the beginning of the previous `Student` record in the file.

Use Case:

This is commonly used in scenarios where you are iterating over records in a file, and you need to modify or re-read a record that was just accessed.

Example:

```c
struct Student {
    char name[50];
```

```
    int id;

    float grade;

};


FILE *file = fopen("students.dat", "rb+");


// Assume we have already read a Student record and now want to modify it

fread(&student, sizeof(Student), 1, file);


// Move the file pointer back to the start of the Student record we just read

fseek(file, -sizeof(Student), SEEK_CUR);


// Write the modified Student record back to the file

fwrite(&student, sizeof(Student), 1, file);
```

In this example, after reading the student record, the file pointer is moved back by `sizeof(Student)` bytes so that the next `fwrite` operation will overwrite the same record.